

Today's content.

→ Max consecutive 1's by

a) Atmost 1 Replace

b) Atmost 1 swap.

→ Count Triplets

→ Josephus problem.

Q: Given a binary array, we can at most replace a single '0' with '1', find the max consecutive 1's we can get.

ex: $1\ 1\ 0\ 1\ 1\ 0\ 1\ 1$ ans = 5.

ex: $0\ 1\ 1\ 1\ 0\ 1\ 1\ 0\ 1\ 1\ 0$ ans = 6.

$\leftarrow 1 \xrightarrow{3}$
 $0 + 1 + 3 = 4$

$3 + 1 + 2 = 6$

$2 + 1 + 2 = 5$

$2 + 1 + 0 = 3$

ex: $1\ 1\ 1\ 1\ 1\ 1$, ans = 6

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

ex: $1\ 1\ 1\ 0\ 1\ 1\ 1\ 1\ 1\ 1\ 0\ 0\ 1\ 1\ 0\ 1\ 1$

$\leftarrow 2 \xrightarrow{2}$
 $3 + 1 + 2 = 6$

$2 + 1 + 4 = 7$

$4 + 1 = 5$

$0 + 1 + 2 = 3$

$1 + 1 + 2 = 5$

ans = 7.

Steps

For every '0'.

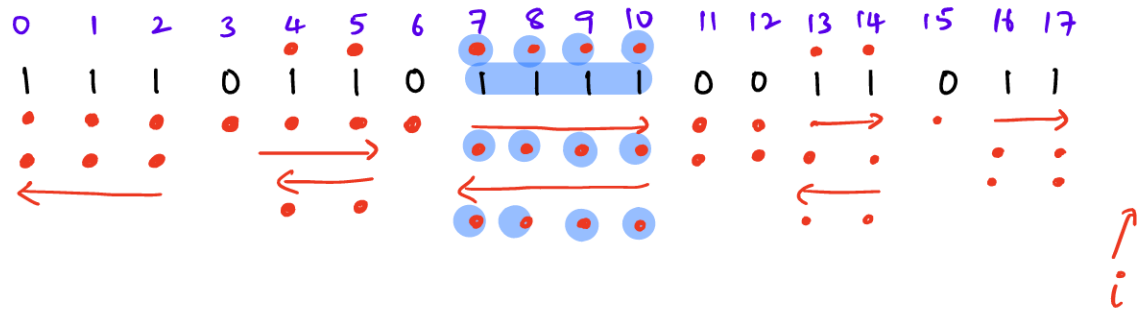
1. count consecutive ones on left $\rightarrow l$
2. count consecutive ones on right $\rightarrow r$
3. ans = max(ans, l+r+1)

Edge case: If all are 1's, then return n.

Code:

```
int findMaxConsOnesByReplace (int ar[N])  
  
    int c = 0  
    for (i = 0; i < n; i++)  
        c = c + ar[i]  
  
    if (c == n)  
        return n  
  
    for (i = 0; i < n; i++)  
        if (ar[i] == 0)  
            int l, r = 0  
            // i's on left.  
            for (j = i + 1; j >= 0; j--)  
                if (ar[j] == 1)  
                    l++  
                else  
                    break  
  
            // i's on right  
            for (j = i + 1; j < n; j++)  
                if (ar[j] == 1)  
                    r++  
                else  
                    break  
  
            ans = max(ans, l + r + 1)  
  
    return ans
```

TC: Discussion.



In worst case you're touching every element 3 times.

$\Rightarrow 3 * n$ iterations.

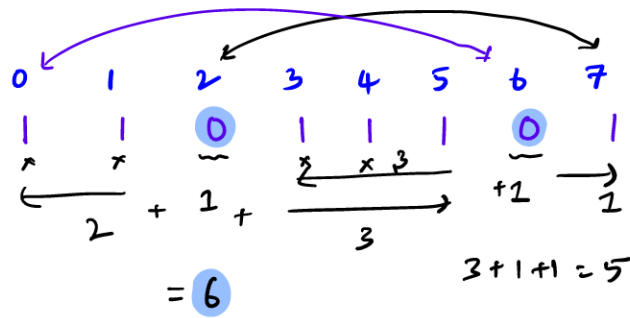
$\Rightarrow TC: O(3n) \Rightarrow O(n)$.

Note: Whenever you've 'if' & 'break' in your loops, dry run & find TC.

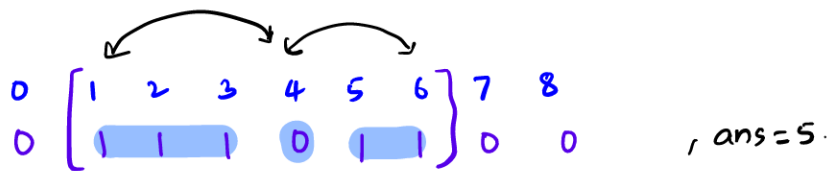
Don't jump in.

20: Given a binary array, we can at most swap a single '0' with '1', find the max consecutive 1's we can get.

ex:



ex:



if there's an extra one, $ans = \max(ans, l+r+1)$

if there's no extra one, $ans = \max(ans, l+r)$

↳ Implies [total no. of ones = $l+r$].

Code:

```
int findMaxConsOnesBySwap (int ar[n])
```

```
    int c = 0
```

```
    for(i=0; i<n; i++)
```

```
        c = c + ar[i]
```

```
    if(c == n)
```

```
        return n
```

```
    for(i=0; i<n; i++)
```

```
        if(ar[i] == 0)
```

```
            int l, r = 0
```

```
            // i's on left.
```

```
            for(j=i+1; j>=0; j--)
```

```
                if(ar[j] == 1)
```

```
                    l++
```

```
                else
```

```
                    break
```

```
            // i's on right
```

```
            for(j=i+1; j<n; j++)
```

```
                if(ar[j] == 1)
```

```
                    r++
```

```
                else
```

```
                    break
```

```
            if(c == l+r)
```

```
                ans = max(ans, l+r)
```

```
            else
```

```
                ans = max(ans, l+r+1)
```

```
    return ans
```

TC: $O(n)$

SC: $O(1)$

10:19:00

10:29:00

10:30:00

3Q:

No. of triplets.

Given $ar[N]$ elements, calculate no. of triplets i, j, k such that $i < j < k$ and $ar[i] < ar[j] < ar[k]$, i, j, k are indices.

Ex:

$ar[5] : [2, 6, 9, 4, 10]$.

ans = 5.

$i < j < k$	$ar[i] < ar[j] < ar[k]$
$0 < 1 < 2$	$2 < 6 < 9$
$0 < 1 < 4$	$2 < 6 < 10$
$0 < 2 < 4$	$2 < 9 < 10$
$0 < 3 < 4$	$2 < 4 < 10$
$1 < 2 < 4$	$6 < 9 < 10$

Ex:

$ar[6] : [4, 1, 2, 6, 9, 7]$
 $i \quad j \quad k$

i	j	k	$ar[i]$	$ar[j]$	$ar[k]$
0	3	4	4	6	9
0	3	5	4	6	7
1	2	3	1	2	6
1	2	4	1	2	9
1	2	5	1	2	7
1	3	4	1	6	9
1	3	5	1	6	7
2	3	4	2	6	9
2	3	5	2	6	7

ans = 9.

Idea 1: Generate all pairs such that $i < j < k$ and check if $arr[i] < arr[j] < arr[k]$.

```
int countTriplets(int arr[N])
```

```
    int count = 0
```

```
    for (i = 0; i < n; i++)
```

```
        for (j = i + 1; j < n; j++)
```

```
            for (k = j + 1; k < n; k++)
```

```
                if (arr[i] < arr[j] && arr[j] < arr[k])
                    count++
```

```
    return count
```

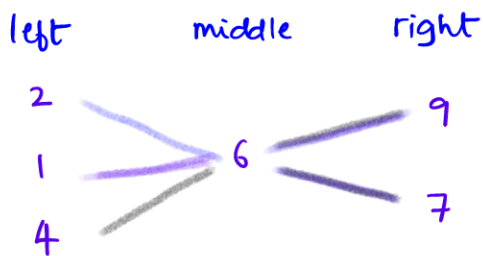
TC: $O(N^3)$

SC: $O(1)$

Can we do better?

Ex: $arr[6] : \{ 4, 1, 2, 6, 9, 7 \}$

hint: how many triplets can you get by keeping index '3' in middle.



we've 6 triplets ($\because 3 * 2$)

$\Rightarrow$ [No. of ele's on left] * [No. of ele's on right]
[ele's smaller than 6] * [ele's greater than 6].

	0	1	2	3	4	5
	4	1	2	6	9	7
l	0	0	1	3	4	4
r	3	4	3	2	0	0
lr	0	0	3	6	0	0

$\{ 4, 1, 2, 6, 9, 7 \}$
↑

, ans = $3 + 6 = 9$.

Idea:

For every $ar[i]$.

(i) Calculate no. of ele's $< ar[i]$ on left $\rightarrow l$

(ii) Calculate no. of ele's $> ar[i]$ on right $\rightarrow r$

$ans = ans + (l * r)$

Code:

```
int countTriplets(int ar[N])
{
    int ans = 0;
    for (i = 0; i < n; i++)
    {
        int l = 0, r = 0;
        // no. of elements < ar[i] on left.
        for (j = 0; j < i; j++)
        {
            if (ar[j] < ar[i])
                l++;
        }
        // no. of elements > ar[i] on right
        for (j = i + 1; j < n; j++)
        {
            if (ar[j] > ar[i])
                r++;
        }
        ans = ans + (l * r);
    }
    return ans;
}
```

0 1 2 3 4 5
{ 4, 1, 2, 6, 9, 7 }
 ↑
 i

TC: $O(N^2)$

SC: $O(1)$

$O(N \log N)$

BST, Segment tree

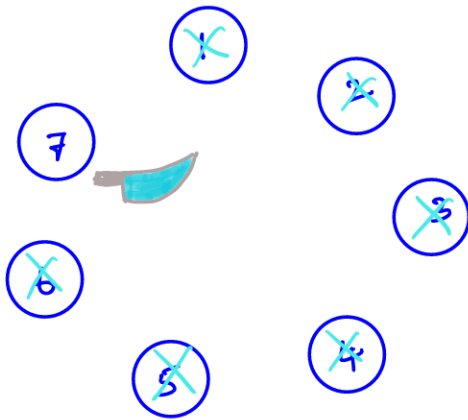
Q4: Josephus Problem. [Squid game in programming].

N people are standing in circle, Person 1 has a knife, he kills next person in clockwise direction and passes on the knife to the next person in the circle.

Everybody repeats this until a single person is alive.

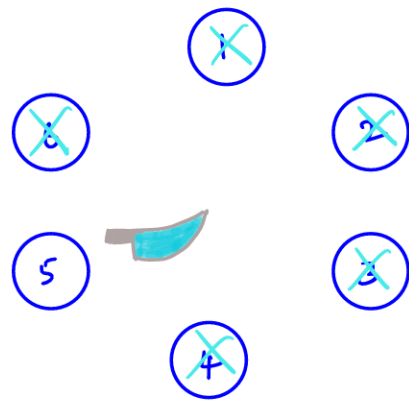
Find the last person standing?

$N=7$.



ans = 7.

$N=6$.

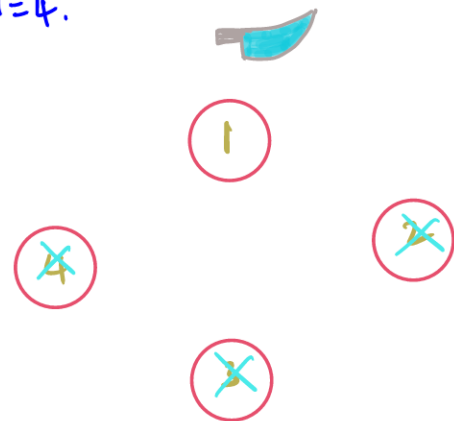


ans = 5.

Observations

1. If n is odd its same.
 2. If n is even its $(n-1)$.
- } X

$N=4$.

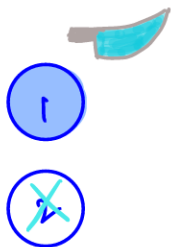


ans = 1.

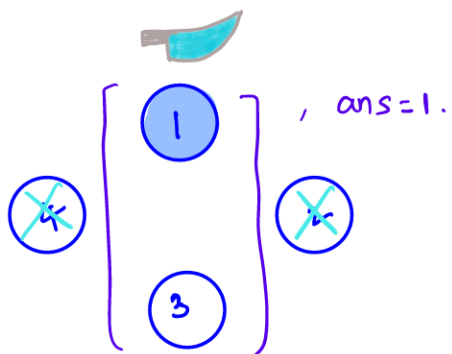
$N=1$



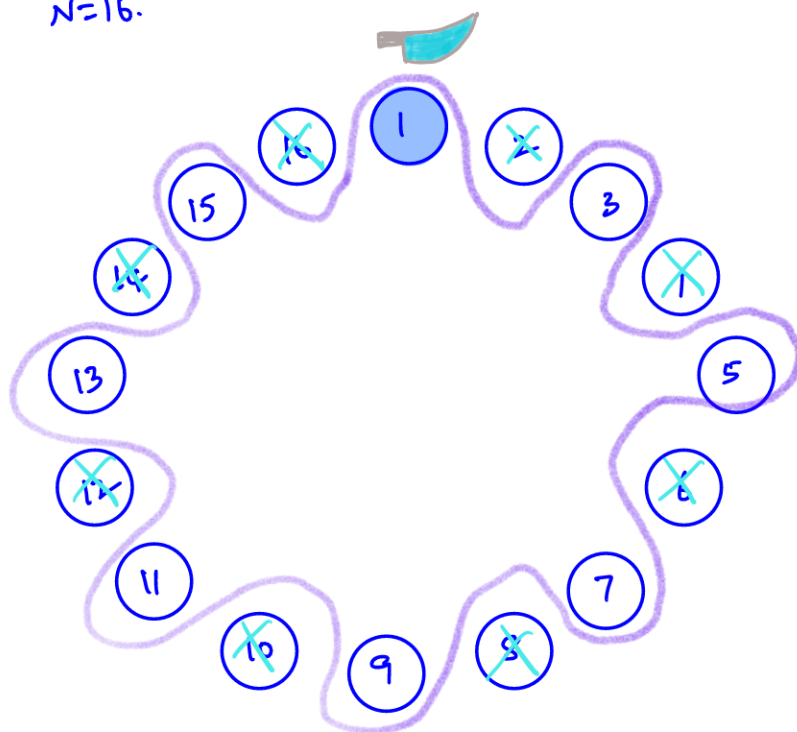
$N=2$



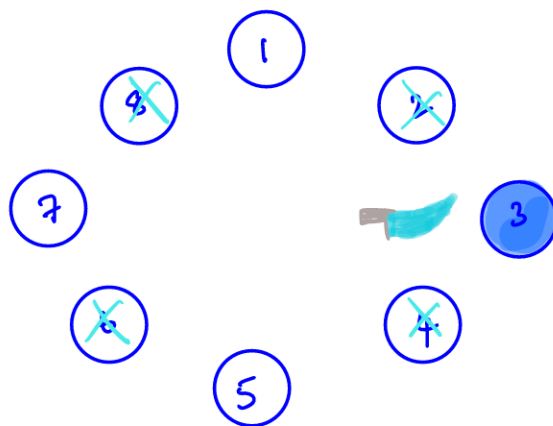
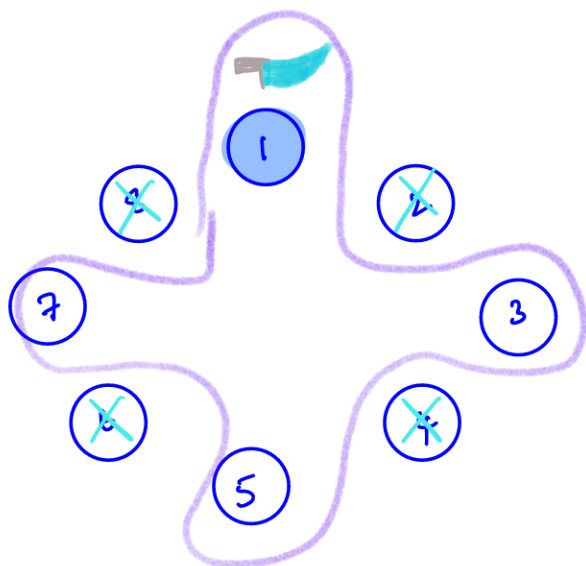
$N=4$



$N=16$



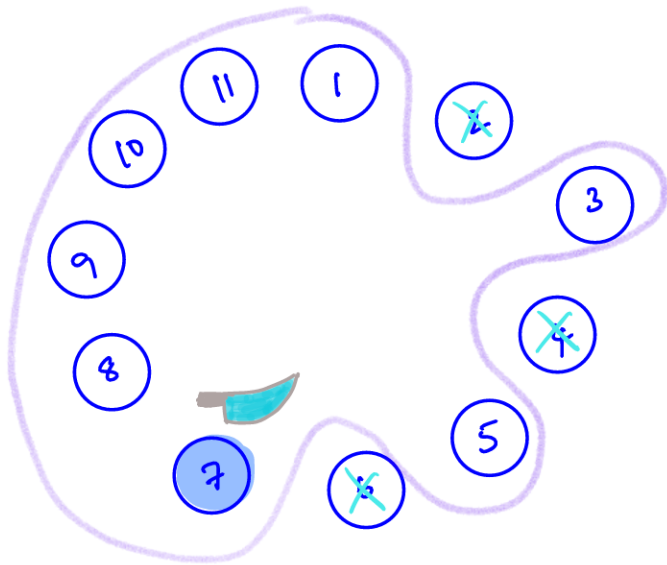
$N=8$



Observation: If n is a power of 2, 1 is the winner.

i.e.: If n is a power of 2, whoever is having the knife at start will be the winner.

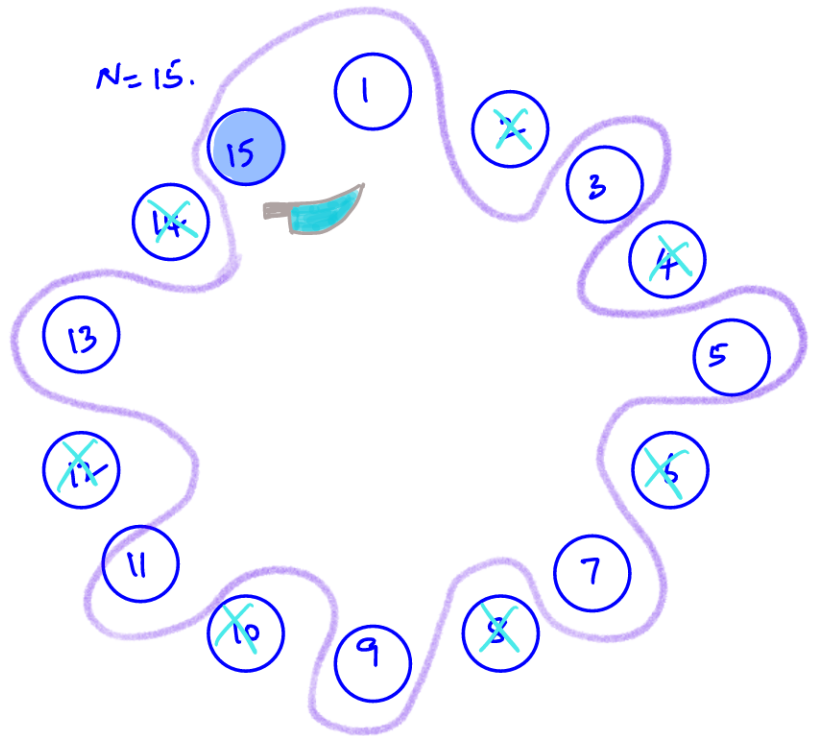
$$N=11, \xrightarrow[1+(3*2)=7]{3 \text{ kills}} 8$$



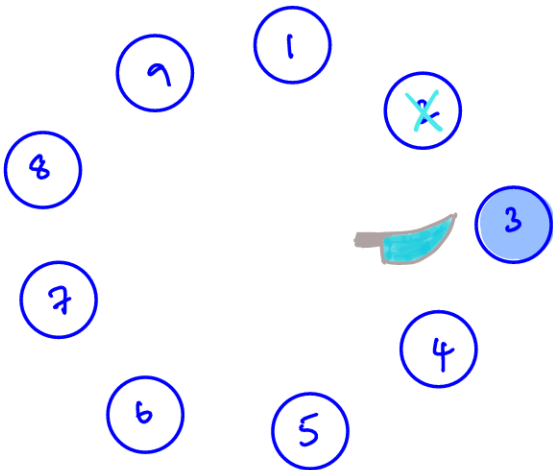
$$1+[7*2]=15 \checkmark$$

$$15 \xrightarrow{7 \text{ kills}} 8$$

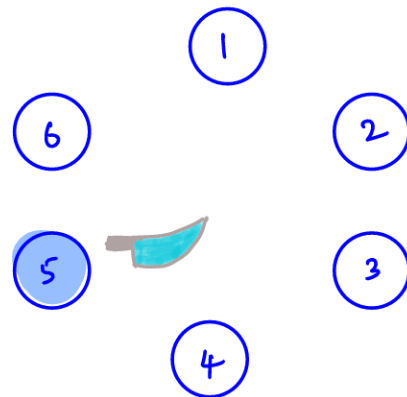
$N=15$.



$$N=9, \xrightarrow[1+(1*2)=3]{1 \text{ kill}} 8$$



$$N=6, \xrightarrow[1+2*2=5]{2 \text{ kills}} 4$$



$$N=100 \xrightarrow[1+(36*2)=73]{36 \text{ kills}} 64$$

Josephus Problem

- You can start anywhere
- You can take 'k' steps.